Procedural Shoot 'Em Up

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Aims

Motivation: Explore the use of procedural generation techniques in the shoot 'em up genre.

Objectives:

- Create a polished, well-tested, exciting shoot 'em up game in Unity.
- Develop a procedural algorithm for generating environments.
- Implement procedural enemies with unique sprites and abilities.
- Create levels and waves procedurally accounting for pacing and difficulty progression.

Design Goal: Tense, challenging gameplay from simple, satisfying systems that evokes an immersive flow state.

Process

Enemy & level generation

- modular sprite textures

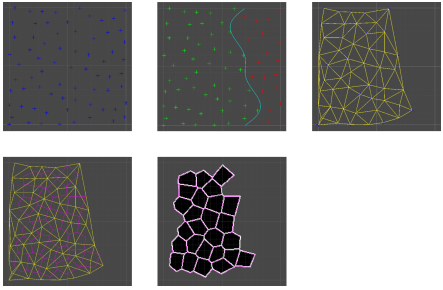


- scriptable objects



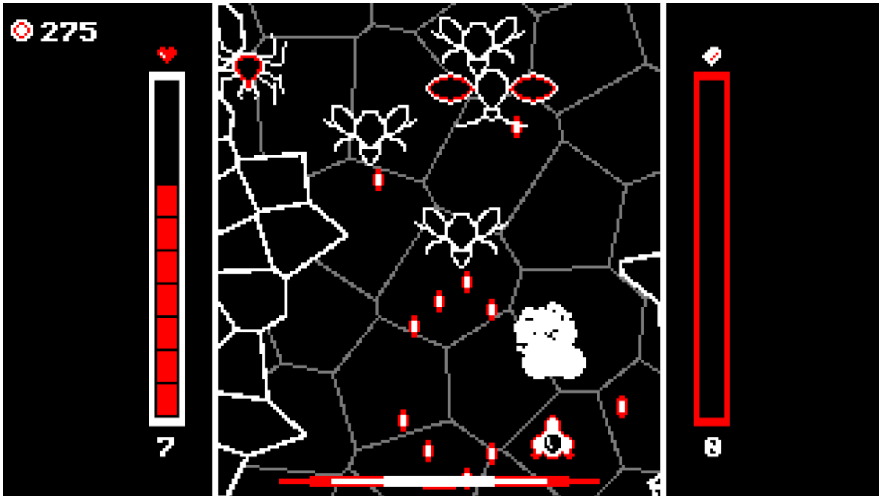
Environment generation

- Poisson disc sampling
- Perlin fractal curve
- Delaunay triangulation / Voronoi partitioning



Result

Instar



Demo & Questions

Onto the demonstration...

Thank you for your attention!
Feel free to ask any questions.